

Homework 1

CENG 113, SP 2026

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Academic Integrity: For all problems **show your work** to receive credit. By submitting your assignment you're confirming that your submission is a product of your own effort.

Box all answers!

Problem 1

Create a plot of the expression

$$0.02946 = \frac{0.556\xi(2-\xi)}{[0.5(1-\xi)]^{1/2}[1.5(1-\xi)]^{3/2}}$$

in rootfinding form (RFF). Use `fsolve` to find all values of ξ which satisfy this expression. Mark these values as circles on your plot. Label your axes. (ANS: $\xi \approx 2, 0.03$)

Problem 2

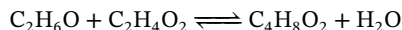
Consider the following system of two equations with two unknowns:

$$\begin{aligned}\xi_1 &= 2.4 - 1.1\xi_2 - \sqrt{(2.4 - 1.1\xi_2)^2 - 1.6(2 - \xi_2)} \\ \xi_2 &= -4 + \sqrt{16 + 4\xi_1(1 - \xi_1)}\end{aligned}$$

Rearrange both equations to rootfinding form, then use `fsolve` to solve both equations simultaneously (*i.e.*, do not substitute one equation into the other, the point is to practice solving systems of equations with `fsolve`). Both ξ_1 and ξ_2 are positive and should be less than 1.

Problem 3

Consider liquid-phase equilibrium between ethanol and acetic acid to form ethylacetate and water according to the reaction

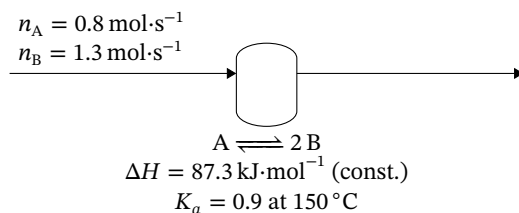


(note that Table D.3 in your textbook incorrectly states the formula of ethanol as $\text{C}_2\text{H}_6\text{O}_2$)

A feed of $1 \text{ kg} \cdot \text{min}^{-1}$ each of ethanol and acetic acid are provided to an equilibrium reactor operating at 75°C . Calculate the conversion of ethanol (as X_A) and the outlet **mass** flow rates of all species if the reactor operates at 250 kPa. (ANS: ethyl acetate $\approx 0.3 \text{ kg} \cdot \text{min}^{-1}$)

Problem 4

Consider the following gas-phase equilibrium reactor:



(4.B) Calculate the outlet mole fractions of all components if the reactor operates at 100°C and 1.2 atm. (ANS: $y_A \approx 0.85$)

(4.C) Plot the conversion of A (as X_A) from 50°C to 300°C and 1.2 atm. Provide a brief (2-3 sentence) explanation of why the curve is shaped the way it is.

Problem 5

In addition to the reaction in the previous problem suppose the gas-phase reaction $2\text{B} \rightleftharpoons \text{C}$ also takes place. This new reaction has an equilibrium constant $K_a = 1.1$ at 90°C with $\Delta H = -62.9 \text{ kJ} \cdot \text{mol}^{-1}$ (constant). Calculate the outlet molar flow rates of all species assuming the reactor operates at 100°C and 1.2 atm. (ANS: $n_B \approx 0.23 \text{ mol} \cdot \text{s}^{-1}$)

(4.A) Estimate K_a at 100°C . (ANS: ≈ 0.03)